Lesson 5.6 Guided Notes

Session 1 · Drawing Nets of Solids

I can draw the net of a solid by unfolding its faces onto a flat surface. · 6.G.A.4

Word / Spanish	Meaning — complete the blank	Example
net / plantilla	A flat _____ that folds into a solid.	an unfolded box
face / cara	One _____ side of a solid.	a cube has 6 faces
prism / prisma	A solid with two matching _____ joined by rectangles.	a box, a tent
pyramid / pirámide	A solid with one base and _____ faces that meet at a point.	square base + 4 triangles

Key idea

Every face of the solid appears _____ in its net.

Edges that fold together must have the same _____.

A rectangular prism net has _____ rectangles in 3 matching pairs.

Worked example

Draw a net for a box 5 cm long, 3 cm wide, and 2 cm tall.

Bottom and top: two _____ cm by 3 cm rectangles.

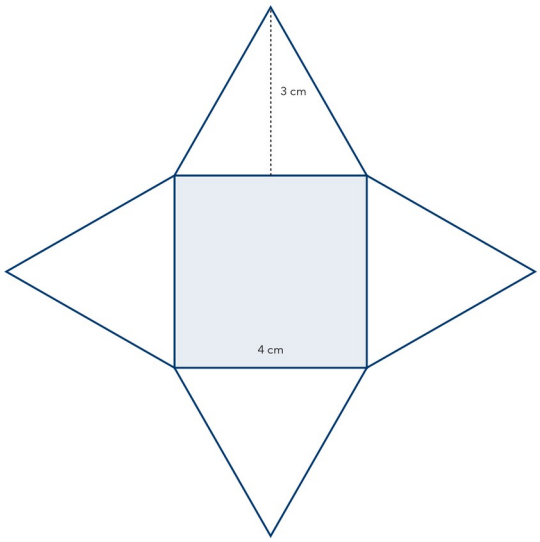
Front and back: two 5 cm by _____ cm rectangles.

Left and right: two _____ cm by 2 cm rectangles.

Answer: _____ rectangles in all

You try

1. This net folds into a square pyramid.
a. Area of the base
b. Area of one triangle



_____

2. How many shapes are in each net?
a. cube: _____
b. triangular prism: _____

_____

Sum it up: To check that a net is correct, I ___ and ___.

_____

Session 2 · Reading Nets and Solids

I can name the solid a net makes by reading the shapes in the net. · 6.G.A.4

Word / Spanish	Meaning — complete the blank	Example
edge / arista	A _____ where two faces meet.	a cube has 12 edges
vertex / vértice	A _____ where edges meet.	a cube has 8 vertices
base / base	The face that _____ a prism or a pyramid.	hexagon base: hexagonal prism

Key idea

A prism has _____ matching bases joined by rectangles.

A pyramid has _____ base and triangles that meet at a point.

Count edges and vertices on the _____ solid, not on the flat net.

Worked example

A net has 1 square and 4 triangles. Name the solid and count its faces, edges, and vertices.

Pieces: 1 + 4 = _____ faces.

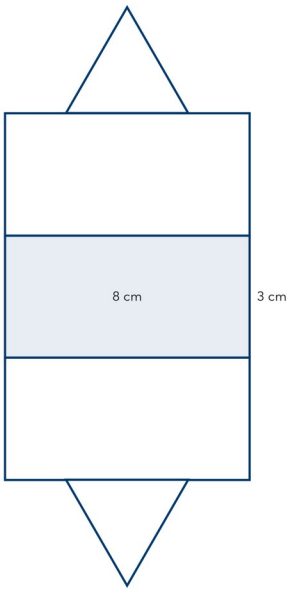
One square base with triangles: a square _____.

Edges: 4 + 4 = _____. Vertices: 4 + 1 = 5.

Answer: square pyramid: 5 faces, 8 edges, _____ vertices

You try

1. Look at the net.
a. Name the solid.
b. Number of faces



_____

2. Name the solid each net makes.
a. 2 octagons + 8 rectangles: _____
b. 1 hexagon + 6 triangles: _____

_____

Sum it up: I can tell a prism from a pyramid by ___.

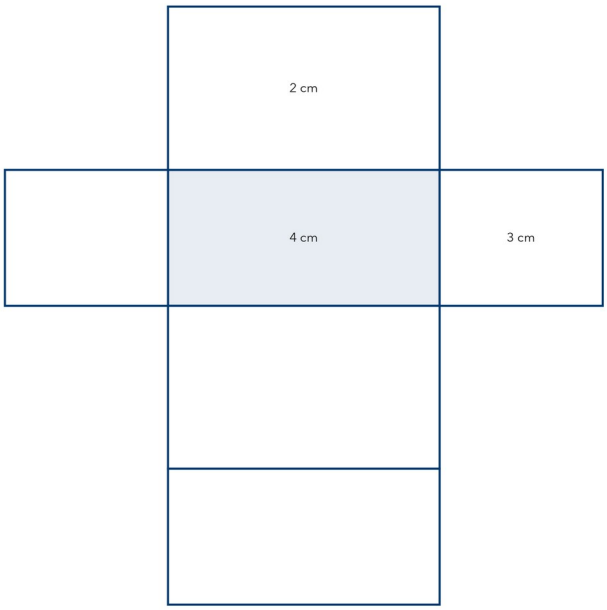
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Session 1 · Model, reason, and explain

Drawing Nets of Solids · 6.G.A.4

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Draw and label a net for a 4 cm by 3 cm by 2 cm rectangular prism. Pair the matching faces and identify which edges must meet.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: treats the right face count as proof a net is valid.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

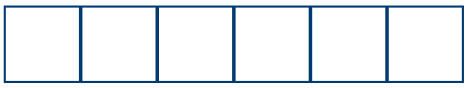
Session 2 · Model, reason, and explain

Reading Nets and Solids · 6.G.A.4

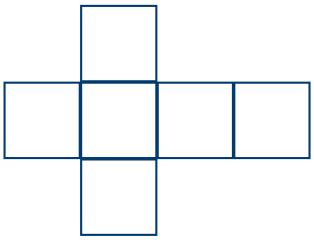
Success: represent the quantities, justify the relationship, and check the result with the correct units.

Compare two arrangements of six congruent squares: a row of six, and four in a row with one above and one below the second square. Which folds into a cube? Explain before testing.

A



B



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: infers a cube from any six squares.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?